



The evolution and decline of traditional water management systems: a case study of Jaipur, India

Kanika Bimrah¹ · Izuru Saizen¹

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Abstract

Since their inception, human civilizations have settled around water resources and developed indigenous water management strategies. However, owing to population growth and rapid urbanization over the last century, these traditional systems have declined. One such example is the case of Jaipur City, which was planned in the eighteenth century. This study examines the evolution and decline of traditional water management systems (TWMS) in Jaipur, one of the earliest systematically planned cities in post-medieval India. Using a combined macro- and micro-level approach, the research integrates historical analysis with primary field surveys of 35 traditional water structures conducted between April and May 2023. The results show a clear transition from decentralized, community-managed systems such as traditional water structures (*stepwells*, *kunds*, and *tankas*) and lakes to centralized, piped water supply networks, driven first by colonial engineering interventions and later by post-independence urban expansion. This research contributes a chronological and spatial understanding of Jaipur's hydrological transformation, identifying key phases of neglect and policy misdirection that led to the loss of traditional water systems. It highlights the urgent need to reintegrate these heritage structures into modern water management frameworks as decentralized recharge systems and cultural assets, essential for improving urban water resilience in arid and semi-arid regions.

Keywords Traditional water management systems · Urbanization · Groundwater Resources · Sustainable Practices

✉ Izuru Saizen
saizen.izuru.4n@kyoto-u.ac.jp

¹ Laboratory of Regional Planning, Graduate School of Global Environmental Studies, Kyoto University, Yoshidahonmachi, Sakyo, Kyoto 606-8501, Japan

Introduction

Research background

Water, a vital element of life, has been an essential resource for the development and sustenance of human civilization since its inception. All great ancient civilizations, such as the Egyptian, Roman, and Indus Valley civilizations, were established around water resources, which helped them survive and flourish (Hosseiny et al. 2021). One of the earliest water management systems developed was in the Indus Valley civilization (~3000–1500 BCE). It is renowned for its advanced water management and drainage systems, which were adapted to the civilization's socio-cultural and socio-economic conditions (Lofrano and Brown 2010). Despite the hot and dry climate and low rainfall in the region, the civilization achieved water sufficiency for its residents (Murthy et al. 2022).

Building upon the ancestral knowledge, the arid and semi-arid regions of western India, particularly state of Rajasthan, further evolved distinctive systems such as stepwells, *kunds*, and *tankas* to cope with scarce and seasonal rainfall (Chandra 2015). These traditional structures not only ensured a reliable supply of water for drinking and agriculture but were also interlinked with religious and cultural practices, reflecting community participation (Livingston 2002). Stepwell (commonly known as *baodis*) were primarily constructed along natural slopes, functioning as both rainwater harvesting structures and access points to the groundwater table. Their design enabled the collection and percolation of rainwater to recharge aquifers, while the flight of orchestrated steps made it easier to access the groundwater even as levels fluctuated seasonally. The closely packed design minimized water loss through evaporation (Azmi 2021). *Kunds*, constructed as early as 1600 AD, are saucer-shaped rainwater catchment systems with sloping sides that direct water towards a central underground well, designed primarily for the storage and distribution of drinking water (Murthy et al. 2022). *Tankas* are traditional underground rainwater harvesting structures, lined with finely polished lime, used for centuries in India to collect and store rainwater for various purposes, including irrigation, drinking, and livestock consumption (Kumawat et al. 2023).

However, owing to the changes in population and the built and natural environments with the shifting climate in the last century, these traditional water management systems have become incomprehensible in their engineered function and societal value (Jiang and Arnold 2023). Over the past century, water consumption has increased sevenfold due to population growth, resulting in a significant decline in per capita water availability and consequently impacting water security (Kılıç, 2020). Water resource planning and management face increasing challenges from conflicting stakeholder demands, population growth, rapid urbanization, anticipated climate change effects on the hydrologic cycle, and the increasing frequency of natural disasters.

Considering the aforementioned challenges, there is a critical need to emphasize on the significance of traditional water management systems and their substantial contribution to achieving the aim of ensuring universal access to safe and affordable drinking water by 2030 (Ramadurai 2023). This research focuses on the importance of traditional systems in today's water management and addresses two critical research questions: What factors contributed to the decline of traditional water management systems, and when did the significant paradigm shift in water management practices take place? The term paradigm shift here

refers to the transition from community-managed traditional water systems to centralized, technology-driven urban water supply models that emerged in the mid-twentieth century. The aim of this study is to thoroughly examine the evolution and decline of traditional water management systems in Jaipur city. By understanding their dynamic role from inception to decline, this research seeks to elucidate the interconnections among demographic development, increasing water demand, and resource overexploitation driven by human activities that have influenced these systems.

The analysis explores how these systems were once integral to the urban fabric of semi-arid regions and how changes in these areas have led to their reduced prominence. Furthermore, this study examines the current water management challenges and opportunities in Jaipur, aiming to integrate ancient wisdom with contemporary strategies for water sustainability and the potential revival of traditional water systems as viable components of current water management solutions.

Introduction to case study area

Located in the semi-arid region of the Indian subcontinent, the walled city of Jaipur is recognized as one of the earliest examples of a systematically planned city in post-medieval India (Hooja 2022). It is situated in the foothills of the Aravalli range, surrounded by hillocks on the northern and eastern sides and plains on the western and southern sides (Livingston 2002). Jaipur, the capital city of Rajasthan, is situated approximately 320 km southwest of Delhi, the capital of India (Jawaid et al. 2017) (Fig. 1). The city has a population of 4.1 million people, making it the tenth largest and most populous urban centers in the country (Macro Trends 2023). The summers in Jaipur are very hot, with temperatures up to 47 °C in May, while winters are extremely cold. The region is characterized by relatively low and erratic precipitation, with the monsoon season spanning from July to September (Tyagi et al. 2011). Jaipur has witnessed irregular but intense rainfall, with an average annual rainfall of 633 mm/year from 1970 to 2020 (Meteorological Centre Jaipur

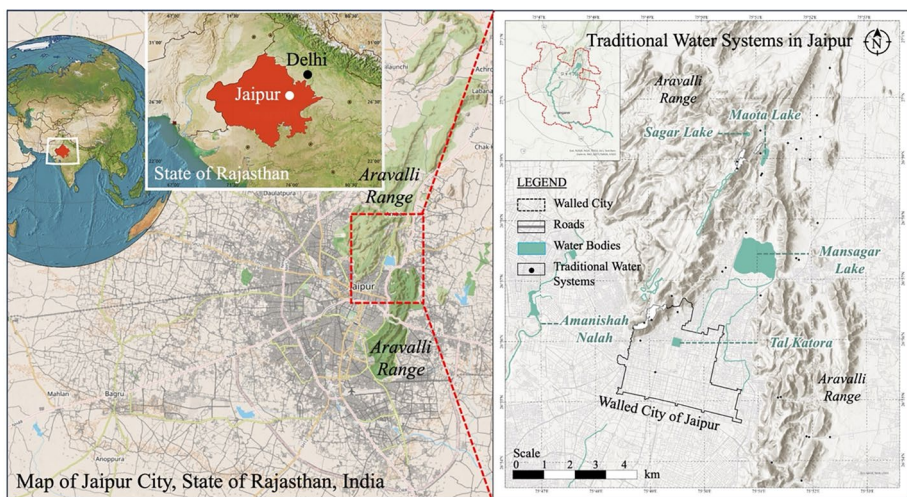


Fig. 1 Location map of case study area: Jaipur, India

2021). Despite being situated near the eastern edge of the Thar Desert and characterized by fluctuating rainfall patterns, the city has been at the forefront of innovative water management systems since its establishment in 1727. Surrounded by the Aravalli range in the north, east, and southeast directions, the city has taken advantage of various catchment areas and developed traditional water systems to sustain its population. Each fort in Jaipur was constructed with a specialized water management system tailored to its specific needs, utilizing efficient rainwater harvesting and groundwater recharge techniques (Shrestha, 2019). During the planning of Jaipur city significant emphasis was placed on the development of traditional water systems within each neighborhood to effectively address the community's water requirements (Khandelwal 2022). This system can be understood as a combination several key components i.e. traditional water management structures that functioned collectively to harvest, store, and distribute water. (i) Traditional water storage structures: such as stepwells (*baodis*), *tankas* and *kunds*, each designed to suit the local topography and community requirements (ii) Urban lakes and water bodies, including *Tal Katora*, Man Sagar, *Maota* Lake, and *Sagar* Lake, which historically served as key water sources and ecological regulators, while *Amanishah Nalah* was the city's sole surface water source at its establishment (iii) Wells within the walled city were distributed across its core, supplying water for domestic and public use.

Research methodology

A comprehensive research methodology was developed for a multidisciplinary research design. It was inspired by the case study method defined by the COSMOS Corporation (Robert 1983) and it incorporated elements of the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines for the identification and selection of traditional water management systems (Page et al. 2021). The aim of this study is to examine the evolution and decline of traditional water management systems in Jaipur, including an extensive case study analysis at the macro and micro levels. The research methodology is structured into a three-step framework comprising (i) definition and design; (ii) preparation, collection, and analysis; and (iii) analysis and conclusion, as shown in Fig. 2.

Stage 1- Define and design: Based on the above-mentioned objective, the first stage of the framework emphasizes on defining the scope and formulating the research design. This study focusses on water resource management at the city-level, with a detailed site-level analysis of traditional water management systems. To identify and select the sites, elements of the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines were adopted. The study follows key PRISMA elements such as identification, screening, and inclusion; to systematically select traditional water management systems in Jaipur from databases and field validation for data analysis. Sites of traditional water systems ($n=45$) were identified in Jaipur City through secondary analysis of data sources such as books, reports, research articles, and the Stepwell Atlas (Dataset prepared as of April 2023). Due to limited access to onsite locations, sites ($n=6$) were excluded during the first screening. In the second screening, sites ($n=4$) were excluded as traditional water management systems have disappeared due to encroachment from the indicated locations, as determined by secondary sources. After the screening process, 35 sites were included for the subsequent stage of data analysis.

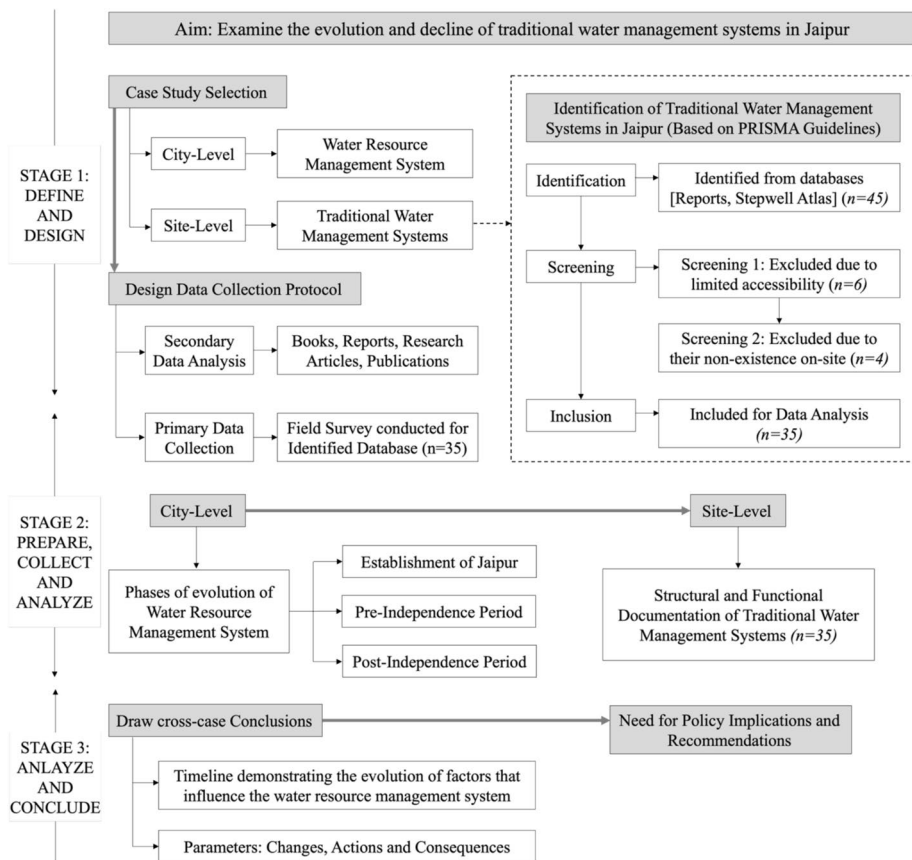


Fig. 2 Research methodology

Stage 2- Prepare, collect and analyse: In the second stage, a systematic analysis of the water resource management system at the city level was conducted. The results were classified into three phases: the establishment of Jaipur, the pre-independence period, and the post-independence period, to provide a comprehensive understanding of the historical evolution of this water management system. Concurrently, a detailed site-level assessment was performed for 35 traditional water management systems in Jaipur City. Primary data were collected through a field survey (see Table 2 Primary Survey Questionnaire) conducted from April to May 2023, which were utilized for structural and functional documentation and analysis.

Stage 3- Analyze and conclude: Cross-case conclusions were drawn from a timeline illustrating the evolution of factors influencing the decline of traditional water management system, categorizing these parameters into changes, actions, and consequences. This study addresses the shortcomings in the planning of the city's water system and delves into the specific aspects of where, how, and when these missteps occurred. The results of this study emphasize the need for policy implications and recommendations with a focus on transforming traditional water management structures to meet the water needs of the local community.

Results

Evolution and decline of traditional water management system

This section examines the chronological development of Jaipur's water management system by categorizing the timeline into three distinct periods: (i) water management during the establishment of Jaipur, (ii) water management in the pre-independence era, and (iii) water management in the post-independence period.¹ The objective of dividing the evolution into these periods was to exemplify the transformation and challenges faced by traditional water systems, particularly stepwells, which were once integral to Jaipur's urban fabric. This study demonstrates how these traditional systems declined as the piped water supply infrastructure developed, signifying dynamic changes in water management techniques over time.

Water management during the establishment of Jaipur

Amer, the capital city of *Kachwaha Rajput*, was established in 967 (Sinha and Rajora 2014). It is surrounded by steep terrain and thus faced challenges in accommodating its growing population. The city faced water security issues, and there was a need for a location with a reliable and sustainable water supply (Borie, Catal  a, et al., 2020). To address the water shortage issues in the hill city of Amer, *Maharaja Sawai Jai Singh II* founded Jaipur in 1727. The new capital of the *Kachwaha* Kingdom was the first planned city in India and was designed to have a population of 50,000 people (Kumar et al. 2015). The ruler rigorously planned the city using his scientific knowledge and transformed natural catchment zones into high-capacity lakes. These lakes, namely, *Man Sagar*, *Tal Katora*, and *Sawai Jai Sagar*, can be seen in the old plans. The cityscape of the walled city of Jaipur featured approximately 820 wells that were used by the residents at that time as well as community wells in public spaces, along with water channels that served the city's gardens and memorials (Borie, Catal  a, et al., 2020). Water structures such as stepwells are important elements of the urban fabric and are located in the city squares of *Badi Chaupar* and *Chhoti Chaupar*. These were connected by underground aqueducts that supplied numerous sources of drinking water at the street level. The first public water supply system consisted of canals fed by water from *Amanishah Nalah* and was established in the mid-1700s for those unable to afford personal wells. The dam on *Amanishah Nalah* was used to channel water to the highest point of the *Chandpole Gate* crest line, distributing it throughout the city, following the natural gradient of the land (Hooja 2022). This canal system, originating from *Chandpole Gate* to *Surajpole Gate*, was connected with stepped tanks and reservoirs first at *Chhoti Chaupar*, then *Badi Chaupar*, and finally *Ramganj Chaupar*. The overflow was directed northeastward toward *Talkatora* and *Man Sagar Lake*, as shown in Fig. 3. Until the 1850s, the primary source was the public water supply system that originated in *Amanishah*, along with groundwater accessed through stepwells and wells.

Water supply management during the pre-independence period

In 1860, during the reign of *Maharaja Sawai Ram Singh II*, the establishment of the Public Works Department (PWD) marked a significant development in Jaipur. In 1867, the Jaipur PWD was succeeded by Major *Samuel Swinton Jacob*, a British military engineer tasked

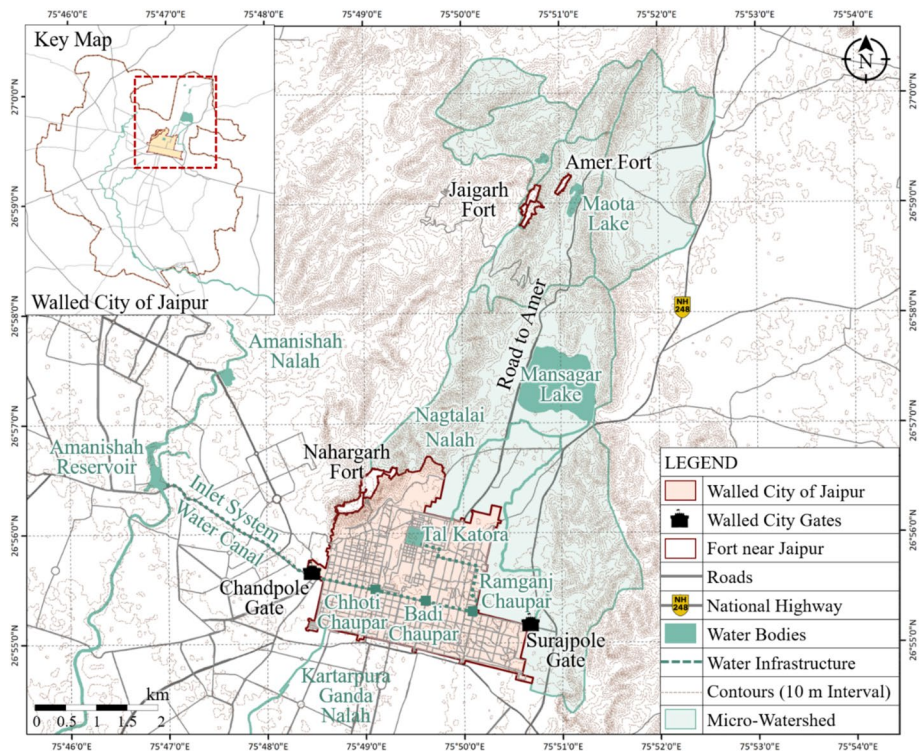


Fig. 3 Water management during the establishment of Jaipur

with constructing roads and water canals across the state. His objective was to connect Jaipur to the surrounding towns and provide irrigation for farmland (Tillotson 2004). Inspired by the principles of British military engineers, *Maharaja Sawai Ram Singh II* initiated the installation of a piped water system in the city in 1868. The stepwells located in the city squares had been a major source of water for the city's inhabitants but were later considered redundant by the introduction of the piped water supply and were judged as unhygienic (Sachdev and Tillotson 2002). As part of this modernization effort, stepwells located in *Chhoti Chaupar*, *Badi Chaupar*, and *Ramganj Chaupar* were filled in and transformed into public squares, disrupting the natural drainage pattern. In 1874, a service reservoir was built across *Amanishah Nalah* that utilized steam engines to pump water, as shown in Fig. 4. The piped water supply was initiated through a 12-inch iron pipe interconnected with branch pipelines that extended into the principal streets. Standposts were strategically installed at each street corner, originating from one reservoir. Water supply was also facilitated by *Hathroi* Fort via a 6-inch delivery line from the same reservoir (Hooja 2022). During the construction of the reservoir, *Amanishah Nalah* exhibited perennial flow; however, over time, its capacity diminished owing to the declining water inflow and significant silting of the dam. To address these challenges, open and tube wells were installed within the reservoir to sustain the water supply (Hooja 2022). The implementation of the *Amanishah* water system marked an exodus from the previous canal and stepwell-based water collection systems. The transition that occurred during this decade marked a fundamental moment in

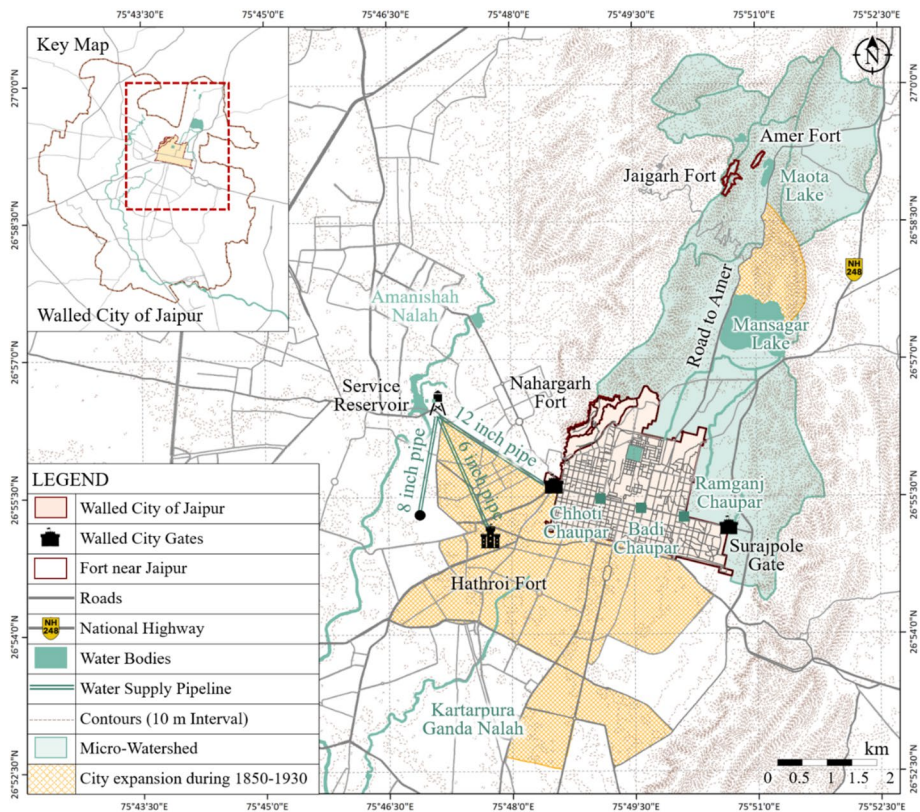


Fig. 4 Water supply management during pre-independence period

the city's history. The adoption of modern piped water systems was crucial for providing a sufficient water supply but came at the expense of diminishing the prominence of traditional water management systems. Several stepwells documented in this study, including Chhoti Chaupar, still exhibit remnants of this transition period, that reflect the transition from traditional water systems to early centralized infrastructure with late nineteenth-century urban reforms. These site-level observations further corroborate archival accounts of how centralized water systems gradually replaced localized catchment-based structures.

The aforementioned water system effectively catered to the city's water demand until the first few decades of the twentieth century. As the population exceeded 100,000, dependence on the water supply scheme increased, and there was a need to augment and expand the water supply system. However, due to multiple droughts in the first two decades of the 1900s and the gradual silting of reservoirs and wells, by the 1920s, the capacity of the *Amanishah* water system was reduced to one-fifth of its original capacity. New sources of water supply were explored, and in 1903, the Ramgarh Dam was constructed 30 km north-east of the city. Water from the dam was initially used for irrigation in the surrounding villages, but with the advent of power generation in 1925, a scheme was developed to transport water to Jaipur. Although the city started receiving water from the Ramgarh Dam in 1931, a household-piped water supply was not common and was only available to well-to-do fami-

lies. Many localities did not have a public distribution system, and reliance on wells was extremely high (Moudgil 2017).

Post-Independence water supply management in Jaipur

Phase 1: Increase in water supply capacity from Ramgarh Dam post-independence (1947–2006): Over the years, the pumping capacity from the Ramgarh Dam witnessed a tremendous increase, rising from 9 MLD in 1953 to 234 MLD in the 1980s due to the implementation of a comprehensive water supply (Roberts et al. 2013). Due to rapid urbanization, concerns regarding the sustainability of Ramgarh Lake began to emerge, and signs of excessive draw-down and occasional complete drying were evident. Anicuts and barriers were constructed across waterways in the catchment area of Ramgarh Lake, which affected the water inflow into the lake (Kumar et al. 2015). In the late 1980s and early 1990s, Ramgarh Lake reportedly remained dry at times, rendering it incapable of contributing additional water supply. By 1999, water had ceased to reach the Ramgarh Dam entirely.

Phase 2: Dependence on ground water resources (2006–2009): The only surface water source that catered to Jaipur City's drinking water requirements was Ramgarh Lake, which began drying up in 1999 and was eventually completely depleted by 2006. Consequently, the city's water supply relies solely on groundwater. Due to the excessive extraction of groundwater, the level plummeted by more than 10 m between 2004 and 2009 (Comptroller and Auditor General of India 2010).

Phase 3: Baisapur water supply project (2009–Present): The Baisapur Water Supply Project began in 1995, when signs of drying up and deterioration of the Ramgarh Dam were evident. However, it was not until 2009 that water from the Baisapur Dam was supplied to the city (Stearns and Agarwal, 2009). It is located approximately 120 km southwest of Jaipur on the Banas River, as can be seen in Fig. 5, was constructed to serve as a regional water reservoir catering to the domestic water requirements of Jaipur, Tonk, and Ajmer districts (Akiko 2015). The total water production in 2010 was 379 MLD, of which 345 MLD was augmented from groundwater sources, while only 34 MLD was supplied by the Baisapur Dam (Jain 2014). Gradually, the capacity of water augmented from the Baisapur Dam increased, and 540 MLD of water was supplied by 2017 (Vyas and Goyal 2020). The per capita supply of water in Jaipur was 85 LPCD, which was much less than the norm of 150 LPCD recommended by the Central Public Health and Environmental Engineering Organization (Comptroller and Auditor General of India 2010).

During this phase, traditional water management systems fell into complete neglect, as the primary focus shifted toward expanding and modernizing the piped water supply network to meet the city's growing demand.

Structural and functional documentation of traditional water management systems

Stepwells (*Baodis*)

To conduct an in-depth micro-level study, a comprehensive field survey was undertaken on the identified traditional water management systems located in Jaipur. This meticulous survey involved on-site assessment and documentation aimed at gathering information related to the structural, functional, and historical aspects of each system. The structural condition

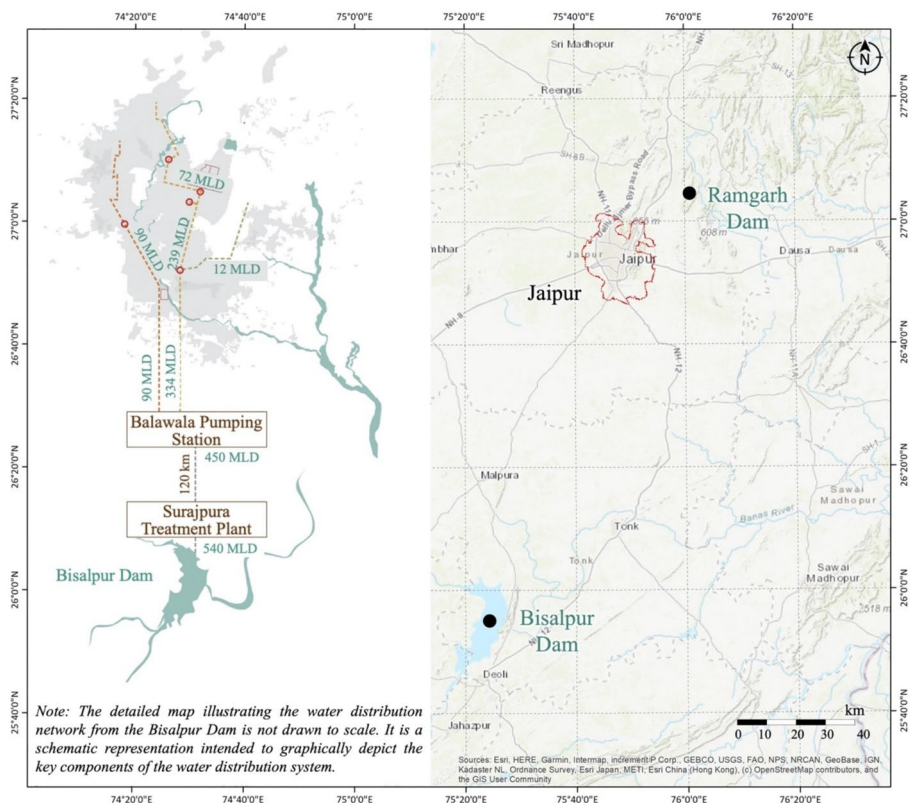


Fig. 5 Water supply from Bisalpur Dam

assessment of the stepwells was integrated into the broader framework of building condition assessment (Yacob et al. 2016). This assessment categorizes stepwells into five distinct categories: (i) good structural condition and in use, (ii) good structural condition but not in use, (iii) dilapidated condition but still in use, (iv) dilapidated and abandoned, and (v) encroached upon. The photographs in Fig. 6 illustrate examples of each of these categories, providing a visual context for the assessment criteria mentioned above.

The functional assessment focused on analyzing the usage of space and water within traditional water management systems, both in the past and in the present.¹ The findings of the structural and functional assessment of traditional systems in Jaipur are documented and mapped in Fig. 7 (Refer Appendix 2: Documentation of Traditional Water Management Systems in Jaipur).

The data collected during the field visit for the traditional water management systems are summarized below. While the full dataset includes numerous parameters, only the subsets relevant to the objectives of this study are presented in the Table 3.

Table 3 Documentation of Traditional Water Management Systems in Jaipur). More than 60 percent of these traditional water management systems are currently either in dilapidated condition or remain non-functional. These architectural marvels, which once stood as vital water sources, have witnessed a transformative shift in their roles over time. Historically,

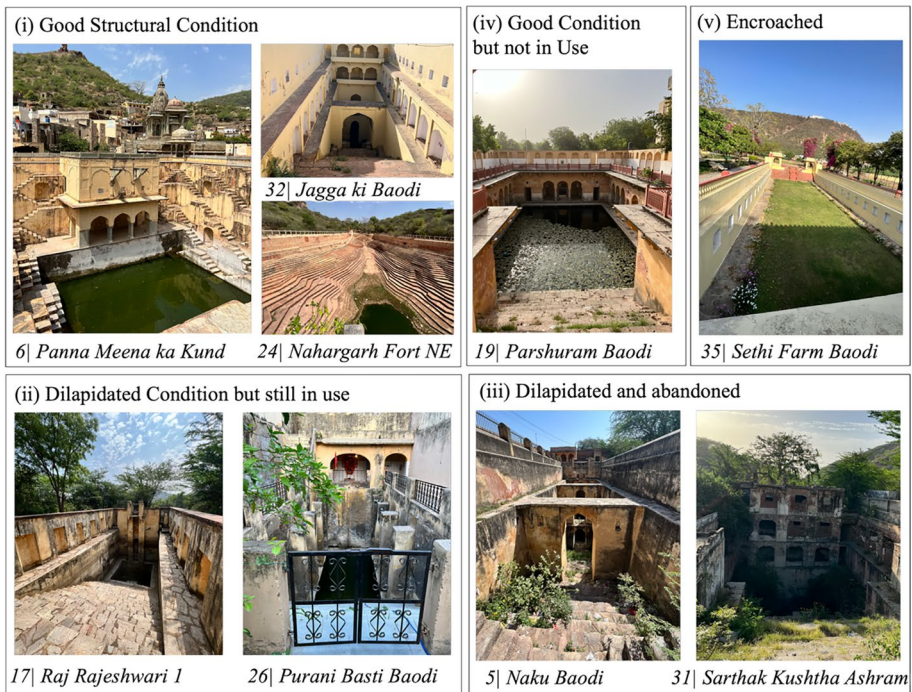


Fig. 6 Photographs of stepwells exemplifying their structural condition

they have provided water for diverse purposes such as agriculture, bathing, construction, drinking, gardening, and religious activities. Over the years, the overexploitation of ground-water resources in Jaipur has led to a substantial decline in both the availability and quality of water in these systems.

At present, the water is predominantly utilized for bathing, gardening, and religious practices. In reality, only 25 percent of traditional water systems continue to fulfill their original purpose of providing water for the local community (Fig. 8a). Regrettably, the current situation exacerbates the issue further, as no system currently hold water suitable for drinking purposes. The absence of drinkable water in these ancient structures not only diminishes their historical significance but also underscores the urgent requirement for sustainable water management strategies to revitalize these architectural gems.

In terms of spatial utilization, approximately 59 percent of the surveyed structures had surrounding areas of religious significance, as shown in Fig. 8b. Historically, these surrounding spaces have served various functions including religious activities, resting areas, open spaces, agricultural land, and recreational activities. The lack of community participation, sense of belonging, and engagement has been attributed to deterioration in the structural condition and utilization of traditional systems. In today's context, the use of these spaces is primarily limited to religious practices, tourism, and recreational activities. While tourism is a modern concept, it parallels the historical role of stepwells as religious and social spaces, reflecting a shift in function from sacred community use to cultural heritage visitation.

Reconstructing the historical decline of Jaipur's traditional water management systems presents considerable challenges due to the lack of precise temporal and spatial data. Field

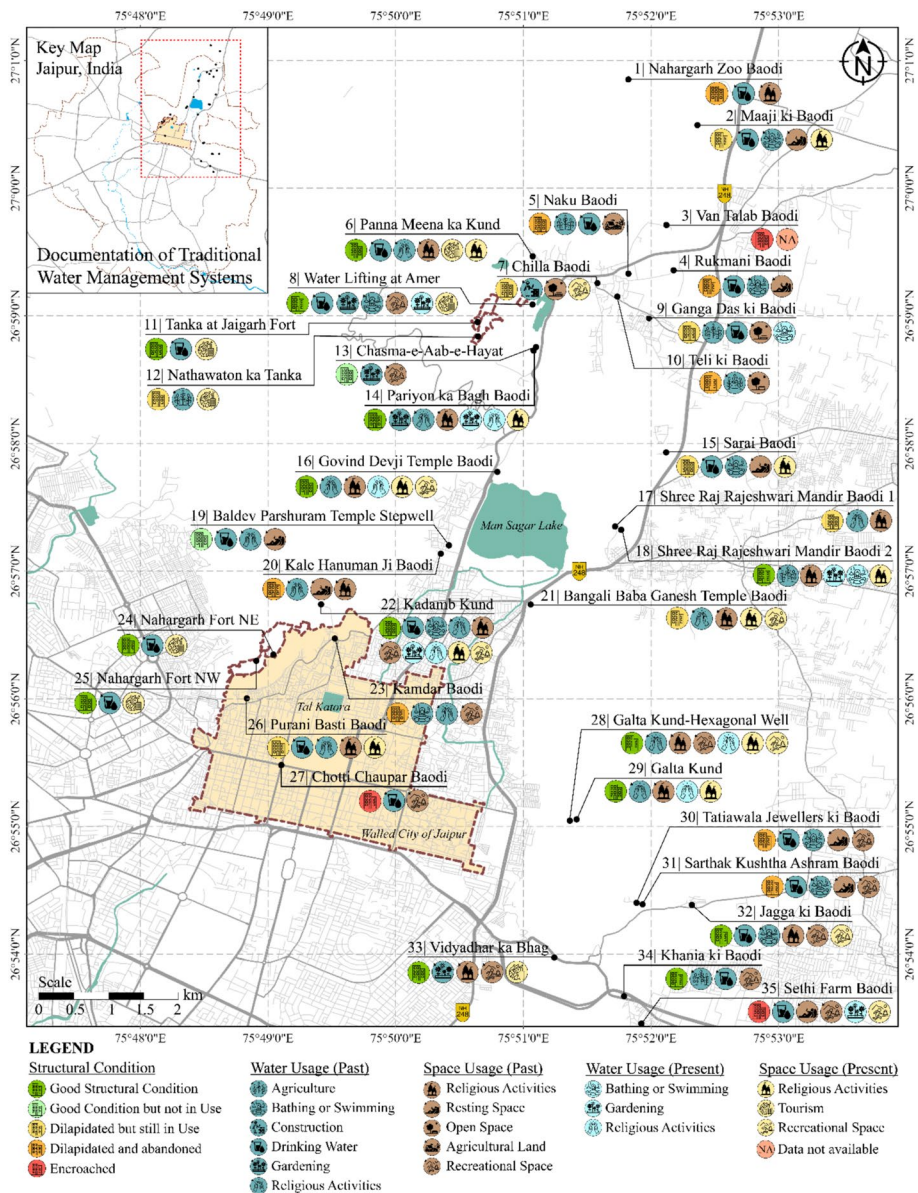


Fig. 7 Structural and functional documentation of traditional water management systems in Jaipur

observations revealed the presence of unauthorized private tubewells near several traditional structures, suggesting localized impacts on groundwater levels and the subsequent loss of functionality in *baodis*. Interviews with community members provided only relative accounts, often expressed without specific time references. Given these limitations, mapping the evolution of parameters became critical to identify the patterns, timing, and

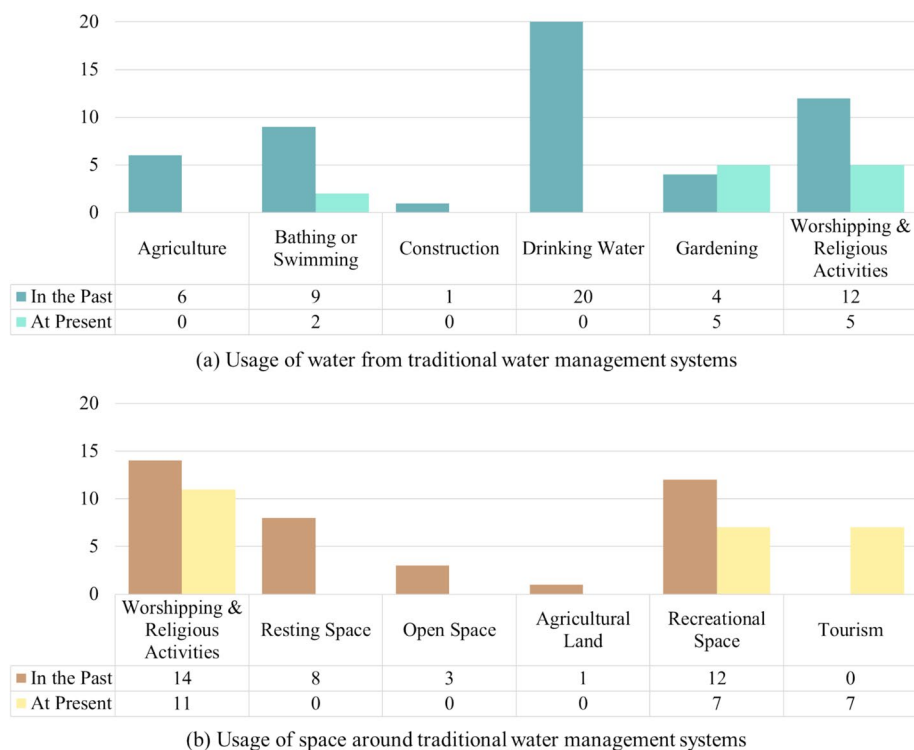


Fig. 8 Change in utilization of traditional water management systems in past and at present

processes through which *baodis* gradually lost their function within the city's changing hydrological and urban context.

Urban lakes (Talabs) and water bodies

There were seven major lakes, namely, *Man Sagar*, *Sagar Lake*, *Tal Katora*, *Maota Lake*, *Santosh Sagar*, *Raja Mal ka Talab* and *Sawai Jai Sagar*, can be seen in the old plans. However, urban expansion has significantly impacted these water bodies over time. The precise timeline of their encroachment remains uncertain; however, only four lakes currently remain in existence within the city of Jaipur (Jaipur Nagar Nigam Heritage 2013).

Sagar Lake

Built in the seventeenth century under *Raja Man Singh of Amer*, *Sagar Lake* served as the primary reservoir for *Amer* and *Jaigarh* Fort. Divided into Upper and Lower *Sagar*, it formed part of an integrated system where overflow from the upper basin recharged surrounding *baodis*, maintaining groundwater during dry periods (Nihal et al., 2021). Beyond its utilitarian role, *Sagar Lake* held ritual significance for idol immersions. Today, despite heritage protection and various restoration efforts, the lake often remains dry and degraded due to reduced inflow and limited restoration.

Man Sagar Lake

Man Sagar Lake supported both the *Jal Mahal Palace* and surrounding ecosystems. Its condition deteriorated after 1962 when sewage inflows from *Brahmpuri* and *Nagtalai nalahs* caused eutrophication, weed growth, and fish decline (S. Sharma and Choudhary, 2021).

Maota Lake

Located at the base of *Amer* Fort, *Maota* Lake historically supplied water to the fort via a hill-fed rainwater collection network. Over time, siltation and stagnation have led to poor water quality and periodic fish kills (Sinha and Rajora 2014).

Tal Katora lake

Constructed in the eighteenth century under Jaipur's Char Bagh planning concept (Dronah 2019), *Tal Katora* functioned as a ceremonial and cultural lake. Once surrounded by gardens and pavilions, it suffered from waste dumping and urban encroachment, leading to severe pollution by the late twentieth century (Gupta et al., 1999).

Amanishah Nalah

Amanishah Nalah, historically a natural drainage corridor connecting the Aravalli foothills to the Dravyavati River, supported irrigation and replenished wells. Over time, it transformed from a seasonal river into a polluted urban drain due to waste disposal and encroachment (Hooja 2022).

Wells within the walled city

Jaipur historically possessed one of the most advanced urban water management systems of medieval India, reflecting its design as a modern trade centre as compared to the conventional fortified settlements typical of Rajput kingdoms. Within its city walls, Jaipur reportedly featured approximately 820 wells (Borie, Françoise, et al., 2020), supplemented by reservoirs, stepwells, tanks, canals, dams, and an aqueduct, collectively fulfilling domestic and livestock water needs. Historical accounts note that only 40 wells provided fresh water, highlighting persistent constraints on potable water access. Affluent residents typically obtained drinking water from select wells known for their potable quality, predominantly located outside the city's limits, reflecting pronounced socio-economic disparities in water resource accessibility (Roy 2006). However, today it is impossible to find any wells of the walled city with water as they have ceased to function effectively. It is either because of contamination with sewage and solid waste or decline in groundwater availability. Residents who live near these wells often recall their former significance as essential community water sources.

Discussion

The multifaceted representation of evolution and the decline of traditional water systems became distinct after examining the historical development of Jaipur's water management system, from its early development as a planned walled city to its existing condition. This section discusses the parameters that contributed to the decline of ancient structures and examines the timeline to determine when the development trajectory deviated from sustainable practices, resulting in the decline of these significant water structures. To thoroughly examine the dynamics of the water management system, a wide range of parameters was considered and classified into systemic changes, infrastructural actions, and their subsequent consequences, as shown in Table 1.

Systemic changes: population growth, urban expansion and water demand

The walled city of Jaipur, which was initially planned to accommodate 50,000 people, has over four million inhabitants as of 2021 (Jaipur Municipal Corporation 2005). Jaipur's spatial extent expanded from 4.81 km² in 1728 AD to 9.6 km² by 1930–31, and further to 40 km² by 1951 (Jawaid et al. 2018) while its built-up area increased from 72.73 km² in 1992 to 283.25 km² in 2020 reflecting rapid post-independence urban sprawl that has intensified surface runoff and reduced groundwater recharge. This exponential growth has been accompanied by a noteworthy evolution in per capita water demand. In the eighteenth century, the city's water supply was intended to provide 20 L of water per capita per day (LPCD) (Misra 2001). Due to the effects of the industrial revolution and rapid urbanization in the next two centuries, per capita water demand increased fourfold to 80 LPCD by 2000 (Amarasinghe et al. 2007). According to the Ministry of Housing and Urban Affairs, the benchmark for urban water supply has been suggested to be 135 LPCD (Ministry of Jal Shakti 2020) and is projected to be 220 LPCD by 2050.

Infrastructural actions

The escalating water demand in Jaipur became a problem that worsened due to the absence of easily accessible surface water sources in proximity. Concurrently, the government authority's interventions, such as increased extraction through groundwater sources, were viewed as critical actions. In 2006, 1340 tube wells operated under the Jaipur water supply, producing approximately 347 million liters of water daily (Jaipur Municipal Corporation 2005). This has resulted in an increase in the city's reliance on groundwater resources as well as their overexploitation. By 2013, despite the city having a water supply from the Bisalpur dam, the number of groundwater extraction sources increased to 1900 (Jain 2014) and it further rose to almost 3252 tube wells by 2022 (Sharma and Mohd 2022).

Consequences: Decline in ground water level and traditional water system

While the demand for groundwater surged over time, the rate of groundwater recharge declined owing to various factors such as urbanization, development of natural drainage patterns, and erratic rainfall. The groundwater level, which was 28.87 m below ground level (MBGL) in 1995 (Singh et al. 2014), has substantially declined to 83 m below ground level

in less than three decades. The decline in the groundwater level and reduced groundwater recharge had a significant impact on the traditional water management systems of Jaipur. The quantity of water available in the stepwells decreased significantly and water quality deteriorated. This twofold problem has resulted in a substantial decrease in the use of these historic structures for water-related purposes, highlighting the adverse consequences of groundwater depletion on traditional water management systems in the region. Based on data obtained from archival sources and the Stepwell Atlas, a total of 45 traditional water structures have been mapped within the city of Jaipur. Over time, the functional use of these structures has significantly declined. Although the government has undertaken restoration efforts time to time, at present only eight of these structures contain water that is actively used by the local residents. The precise timeline of the encroachment of urban lakes remains uncertain; however, only four lakes currently remain in existence within the city of Jaipur. At the time of Jaipur's establishment, approximately 820 wells were located within the walled city. However, only 68 are reported to remain today. Temporal data regarding the stepwells, however, is not available.

This twofold problem has resulted in a substantial decrease in the use of these historic structures for water-related purposes, highlighting the adverse consequences of groundwater depletion on TWMS in the region. The temporal data of all the key parameters discussed above that influence Jaipur's Water Management System is summarized in the Table 1 Evolution of parameters effecting Jaipur's water management system.

Figure 9 elaborates on this discussion further, as it defines the relationships and interactions among the above-mentioned parameters, providing a comprehensive illustration for understanding the intricate network of changes, actions, and their subsequent effects on Jaipur's water management system. Based on the analysis, it can be emphasized that the period between 2000 and 2009 marked a critical juncture, during which the decisions and actions taken by government entities had a profound impact on Jaipur's water resources. During this time, unauthorized construction proliferated within the catchment area of Ramgarh Dam, coupled with significant oversight in forecasting future water demands and a surge in the drilling of numerous tube wells, leading to the overexploitation of groundwater resources. As a result, not only was groundwater availability threatened by this unsustainable exploitation, but traditional water management techniques, such as stepwells in Jaipur, also began to disappear.

Conclusion

The results of this case study, which is based on in-depth examination of 35 ancient water structures, highlight the interplay between demographic shifts and governmental interventions as well as their effects on traditional water management. These findings offer valuable insight and implications for policymakers, urban planners, and researchers striving to identify sustainable water management solutions in similar regions worldwide. The results of this study highlight the importance of taking proactive measures to address the challenges posed by urbanization and development pressures while preserving traditional water management practices. As population growth and water demand surged, traditional water systems, which once served as the backbone of water management in the region, proved inadequate to meet the expanding needs of communities. Consequently, as people began

Table 1 Evolution of parameters effecting Jaipur's water management system

Year	Systemic changes			Infrastructural actions			Consequences			
	Population	Built-up area (sq. km.)	Water demand (MLD)	Water supply SW (MLD)	GW Extrac- tion sources (Number)	Volume of GW extracted (MLD)	Ground water level (MBGL)	Tradi- tional water structures	Urban Lakes	Wells within walled city
1727	50,000 ^a	7.09 ^g	1.00 ⁱ	—	—	—	—	45 ^s	7 ^a	820 ^a
1825	60,000 ^a	—	2.40 ⁱ	—	—	—	—	—	—	—
1871	137,000 ^b	—	5.48 ⁱ	—	—	—	—	43 ^t	—	—
1881	142,578 ^b	—	5.70 ⁱ	1.45 ^m	—	—	—	—	—	—
1891	158,905 ^b	—	6.35 ⁱ	3.33 ^m	—	—	—	—	—	—
1901	160,167 ^b	—	6.41 ⁱ	3.13 ^m	—	—	—	—	—	—
1911	137,098 ^b	—	5.48 ⁱ	2.65 ^m	—	—	—	—	—	—
1921	120,207 ^b	—	4.81 ⁱ	—	—	—	—	—	—	—
1931	144,179 ^c	—	5.77 ⁱ	4.31 ⁿ	—	—	—	—	—	—
1941	175,810 ^c	—	7.03 ⁱ	5.18 ⁿ	—	—	—	—	—	—
1951	291,130 ^c	—	23.29 ^j	9.00 ⁿ	—	—	—	—	—	—
1961	403,444 ^c	—	32.27 ^j	—	—	—	—	—	—	—
1971	615,258 ^c	—	49.22 ^j	27.00 ⁿ	—	—	—	—	—	—
1981	977,165 ^c	—	131.91 ^k	45.00 ⁿ	—	108 ^c	—	—	—	—
1991	1,458,438 ^c	—	196.89 ^k	—	—	153 ^c	—	—	—	—
1992	—	72.73 ^h	—	—	—	—	—	—	—	—
1995	—	86.82 ^h	—	—	—	209 ^c	25.80 ^c	—	—	—
2001	2,374,000 ^d	115.00 ^h	356.10 ^l	—	—	300 ^c	30.14 ^c	—	—	—
2002	—	119.16 ^h	—	—	—	—	—	—	—	—
2003	—	120.10 ^h	—	—	—	—	—	—	—	—
2004	—	122.13 ^h	—	—	—	—	—	21 ^d	—	—
2005	2,770,000 ^d	133.85 ^h	415.50 ^l	—	1340 ^d	347 ^d	35.85 ^c	—	—	—
2006	—	142.16 ^h	—	—	1350 ^p	—	—	—	—	—
2007	—	147.17 ^h	—	—	1646 ^p	382 ^j	—	—	—	—
2008	—	158.89 ^h	—	—	1682 ^p	395 ^j	—	15 ^u	—	—

Table 1 (continued)

Year	Systemic changes			Infrastructural actions			Consequences		
	Population	Built-up area (sq. km.)	Water demand (MLD)	Water supply SW (MLD)	GW Extraction (Number)	Volume of GW extracted (MLD)	Ground water level (MBGL)	Traditional water structures	Urban Lakes within walled city
2009	—	162.37 ^h	—	—	1857 ^p	344 ^j	—	—	—
2010	—	167.29 ^h	—	34.00 ^e	1908 ^p	345 ^j	47.49 ^c	—	—
2011	3,073,350 ^e	182.74 ^h	461.01 ^l	—	1658 ^q	—	50.50 ^r	—	—
2012	—	208.29 ^h	—	—	—	—	—	—	—
2013	—	231.97 ^h	—	331.36 ^o	1300 ^e	89 ^b	56.00 ^r	—	68 ^u
2014	3,112,000 ^e	233.93 ^h	466.80 ^l	339.30 ^o	—	99 ^e	59.00 ^r	—	—
2020	—	283.25 ^h	—	—	—	—	—	—	—
2021	4,008,000 ^f	—	601.20 ^l	432.84 ^o	—	—	80.00 ^r	—	—
2022	4,107,000 ^f	—	616.05 ^l	445.12 ^o	3252 ⁱ	—	83.00 ^r	8 ^v	4 ^a

MLD; Million liters per day, MBGL, Meters below ground level

—Represents that 'no data is available' for that parameter for that year

Source: ^a(Borie, Cataláa, et al., 2020), ^b(Roberts et al. 2013), ^c(Singh et al. 2014), ^d(Jaipur Municipal Corporation 2005), ^e(Jain 2014), ^f(Macro Trends 2023), ^g(Dronah 2019), ^h(European Space Agency (ESA), 2023), ⁱ Calculated by author based on standards mentioned by (Misra 2001), ^j Calculated by author based on standards mentioned by (CPHEEO (Amarasinghe et al. 2007), ^kCalculated by author based on standards mentioned by (CPHEEO 1980), ^l Calculated by author based on standards mentioned by (CPHEEO 1999), ^m(Public Works Department 1915), ⁿ(Kumari and Sharma 2023), ^o(Sharma and Mohd 2022), ^p(Comptroller and Auditor General of India 2010), ^q(ISET and CEDSI 2011), ^r(Rukshar et al. 2023), ^sData compiled by author based on (Atlas of Stepwells, 2016), ^t(Hooja 2022), ^u(Jaipur Nagar Nigam Heritage 2013), ^vPrimary Survey conducted by Author April–May 2023

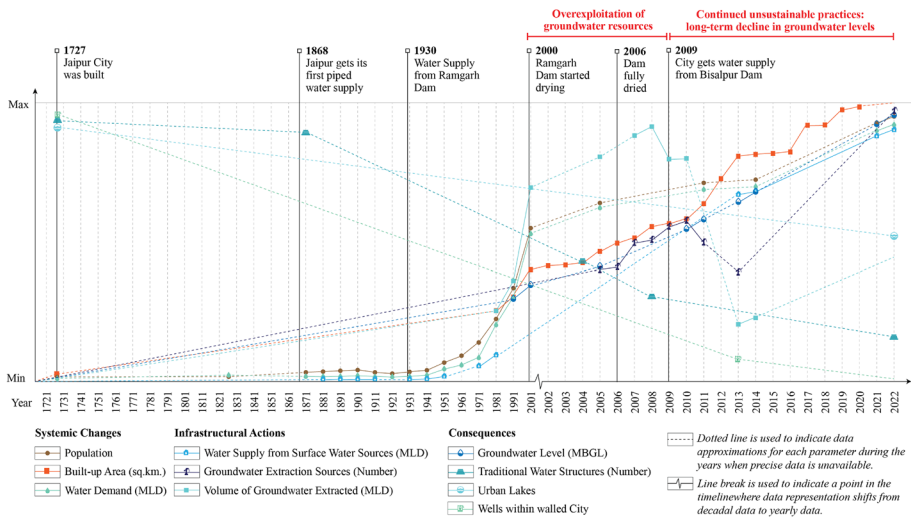


Fig. 9 Evolution of parameters effecting Jaipur's water management system

seeking alternative water supply methods, British representatives introduced a piped water supply to Jaipur City. Additionally, the British colonial administration's policies and infrastructure investments favored centralized water distribution systems, further marginalizing traditional decentralized approaches. This marked a significant departure from reliance on indigenous water management practices, altered the physical infrastructure of water distribution, and transformed the socioeconomic dynamics surrounding water access and usage.

The state plays an important role in maintaining and regulating common water resources. There is a hierarchy of water systems, including local wells, water infrastructure, and lakes. It was maintained by the state (or king), which collected taxes and revenue to maintain these systems. Mishra claims that "[t]he people of Rajasthan did not entrust the organization of such boundless [water] work to either the central or federal government, nor even to what in today's world is termed as the private sector" (Levi and Mishori 2015). Mishra's statement is substantiated through the findings of this research, which clearly demonstrate the shortcomings of water supply management in Jaipur in recent decades. The analysis revealed that both local authorities have consistently failed to effectively plan and manage the city's water supply because of the lack of foresight in infrastructure development and the inability to enforce regulations, leading to unauthorized groundwater connections and rampant overexploitation of water resources. Such mismanagement not only has strained Jaipur's water supply systems but has also endangered long-term water security for its residents. These challenges resonate with Mishra ideologies and highlight the need to adopt more community-based sustainable water management practices again.

While the capacity of groundwater extraction in recent years has been reduced compared to the early twenty-first century, the overexploitation of groundwater resources from 2000 to 2009 has significantly depleted groundwater levels. Unless the government implements robust measures to replenish groundwater in the near future, revitalization efforts will face considerable challenges. This decline in groundwater levels not only affects the availability of water for existing needs but also undermines the historical significance and functional-

ity of historic systems. These structures were originally designed to tap into groundwater reserves and provide reliable water sources for communities. However, the depletion of groundwater owing to over-extraction has caused many stepwells to become non-functional and insufficient for their intended purposes. Furthermore, in the context of traditional water management systems, specifically in Rajasthan, periodic desilting has proven effective in reviving defunct structures and improving their ecological and functional performance (Agarwal and Narain, 1999).

Policy interventions aimed at conserving traditional water management structures are needed. Policymakers must recognize the potential of these traditional systems in contemporary water conservation strategies and how they can be transformed into effective tools for groundwater recharge. Furthermore, incorporating stepwells into the broader framework of water conservation can help alleviate some of the pressure on modern water-supply systems by providing a decentralized method of water replenishment. Community engagement initiatives can raise awareness of the benefits of such integration, fostering a culture of conservation and a sense of belongingness for historical water management practices. The combined effort of local authorities and community involvement is important for the successful revival and utilization of traditional water management structures as functional assets in battling water scarcity.

Table 2 Primary survey questionnaire**1. General information**

Structure name		GPS waypoint	
Date of survey		Structure ID	IN000475 (1)
Location	Latitude:	Longitude:	
Construction year		Renova- tion Year	☐ Not done
Ownership status	☐ One family ☐ Group of families ☐ Community ☐ ASI ☐ Local government		
☐ Others			
Surrounding area	☐ Dense urban development (Old/ New) ☐ Open space ☐ Encroachment		

2. Structural information

Dimensions (in mts.)	Length:	Breadth:	Height:
Type of structure	☐ Open ☐ Closed		Vernacular architecture
			☐ Yes ☐ No
Shape of structure	☐ Helical ☐ Rectangular ☐ Swastik ☐ L-shaped ☐ Asymmetrical ☐ Others		
Typology of structure	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 Number of entrances ☐ Others		
Construction material	☐ Stone Masonry ☐ Rubble ☐ Brick ☐ Others		
Visual characteristics	☐ Plain facades ☐ Lightly decorated with sculptures ☐ Highly decorated with sculptures		

Water quantity mea-
surement technique
Variation in lighting
level

3. Functional information

At the time of construction	At Present
Usage of water	Usage of water
Usage of space	Usage of space

4. Existing condition

Presence of water	☐ Yes ☐ No If yes, depth of water (in mts.)
Structural condition	☐ Good structural condition ☐ Dilapidated but still in use ☐ Dilapidated and abandoned ☐ Good condition but abandoned ☐ Encroached

Reason for structural
condition

Management authority

Conservation measures	☐ Water filtration ☐ Separate sewage Drain ☐ Bio-retention area ☐ Recharge through injection well ☐ Filter drains and trenches ☐ Water treatment ☐ Infiltration Basin ☐ bioswales ☐ Wetlands ☐ Porous pavements ☐ Others If No, What type of conservation mea- sures are feasible
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Table 3 Documentation of traditional water management systems in Jaipur

S.No	Stepwell Name	Structural condition	Usage of water (In the Past)	Usage of water (At Present)	Usage of space (In the Past)	Usage of space (At Present)
1	Nahargarh Zoo Baodi	Dilapi-dated and abandoned	Drinking	None	Religious	None
2	Maaji ki Baodi	Dilapidated but still in use	Drinking, Bathing	None	Resting	Religious
3	Van Talab Baodi	Encroached	Data not available	None	Data not available	None
4	Rukmani Baodi	Dilapi-dated and abandoned	Drinking, Bathing	None	Resting	None
5	Naku Baodi	Dilapi-dated and abandoned	Agriculture, Drinking	None	Agriculture	None
6	Panna Meena ka Kund	Good structural condition	Drinking, Religious	None	Religious	Tourism, Religious
7	Chilla Baodi	Dilapidated but still in use	Construction	None	Open Space	Recreational
8	Water Lifting at Amer	Good structural condition	Drinking, Gardening, Bathing	Gardening	Recreational	Tourism, Religious
9	Ganga Das ki Baodi	Dilapidated but still in use	Agriculture, Drinking	Bathing/ Swimming	Open Space	None
10	Teli ki Baodi	Dilapi-dated and abandoned	Agriculture	None	Open Space	None
11	Tanka at Jaigarh Fort	Good structural condition	Drinking	None	None	Tourism
12	Natha-waton ka Tanka	Dilapidated but still in use	Agriculture	None	None	Tourism
13	Chasma-e-Aab-e-Hayat	Good condition but not in use	Gardening	None	Recreational	None
14	Pariyan ka Bagh Baodi	Good structural condition	Gardening, Religious	Gardening, Religious	Religious	Religious
15	Sarai Baodi	Dilapidated but still in use	Drinking, Bathing	None	Resting	Religious
16	Govind Ji Temple Baodi	Good structural condition	Religious	Religious	Religious	Recreational, Religious
17	Shree Raj Rajesh-wari Mata Ji Mandir Baodi 1	Dilapidated but still in use	Religious	None	Religious	None

Table 3 (continued)

S.No	Stepwell Name	Structural condition	Usage of water (In the Past)	Usage of water (At Present)	Usage of space (In the Past)	Usage of space (At Present)
18	Shree Raj Rajeshwari Mata Ji Mandir Baodi 2	Good Structural Condition	Agriculture	Garden- ing, bathing/ Swimming	Religious	Religious
19	Baldev-Parshuram Temple Stepwell	Good condition but not in use	Drinking, Religious	None	Resting	None
20	Kale Hanuman Ji Mandir Baodi	Dilapidated and abandoned	Religious	None	Religious, Resting	None
21	Ban-gali Baba Ganesh Temple Baodi	Dilapidated but still in use	Religious	None	Religious	Recreational, Religious
22	Kadamb Kund	Good structural condition	Drinking, Bathing, Religious	Gardening, Religious	Religious, Recreational	Recreational, Religious
23	Kamdar Baodi	Dilapidated and abandoned	Bathing, Religious	None	Recreational	None
24	Nahargarh Fort NE Stepwell	Good Structural condition	Drinking	None	None	Tourism
25	Nahargarh Fort NW Stepwell	Good structural condition	Drinking	None	None	Tourism
26	Purani Basti Baodi	Dilapidated but still in use	Drinking, Religious	None	Religious	Religious
27	Choti Chaupar Baodi	Encroached	Drinking	None	Recreational	None
28	Galta Kund—Hexagonal Well	Good structural condition	Religious	Religious	Religious, Recreational	Recreational, Religious
29	Galta Kund	Good structural condition	Religious	Religious	Religious	Religious
30	Tatiwala Jewellers ki Baodi	Dilapidated and abandoned	Drinking, Bathing	None	Resting, Recreational	None
31	Sarthak Kushtha Ashram Baodi	Dilapidated and abandoned	Drinking, Bathing	None	Resting, Recreational	None
32	Jagga ki Baodi	Good structural condition	Drinking, Bathing	None	Religious, Recreational	Recreational

Table 3 (continued)

S.No	Stepwell Name	Structural condition	Usage of water (In the Past)	Usage of water (At Present)	Usage of space (In the Past)	Usage of space (At Present)
33	Vidyadhar ka Bagh	Good structural condition	Gardening	None	Religious, Recreational	Tourism
34	Khania ki Baodi	Good structural condition	Agriculture, Drinking	None	Recreational	None
35	Sethi Farm Baodi	Encroached	Drinking	Gardening	Resting, Recreational	Recreational

Appendix 1: Primary survey questionnaire

The primary survey questionnaire was designed to document the traditional water management structures in Jaipur City. It was structured into four sections to gather data on the structural and functional aspects of these systems.

Appendix 2: Documentation of traditional water management systems in Jaipur

The data collected during the field visit for the traditional water management systems are summarized below. While the full dataset includes numerous parameters, only the subsets relevant to the objectives of this study are presented in the Table 3.

Appendix 3: Evolution of parameters effecting Jaipur's water management system

To understand the evolution and decline of traditional water management system in Jaipur, change in parameters such as population, water demand, ground water extraction sources, ground water extracted, ground water level and water usage of stepwells over the years was identified based on various secondary sources as shown in Table 1.

Author contributions This research paper is a collaborative effort between both authors. The research methodology and data analysis parts were mostly contributed to by Saizen Izuru, who also ensured the study's rigorous design and interpretation of its conclusions. In addition to being mostly in charge of data collecting, documenting, and mapping, Kanika Bimrah was also helpful in gathering and arranging the field data. Both writers worked together to construct the discussion and conclusion sections, which represent their collaborative analysis and synthesis of the research findings.

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Data availability Some or all primary data collected during field survey in April–May 2023, that support the findings of this study are available from the corresponding author upon reasonable request.

Declarations

Conflict of interest None of the authors have any financial or non-financial interests that are directly or indirectly related to the work submitted for publication. The authors have no relevant financial or non-financial interests to disclose.

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Kanika Bimrah is a PhD Graduate from the Graduate School of Global Environmental Studies, Kyoto University, Japan. She holds a degree in Urban Planning from the School of Planning and Architecture, New Delhi, and a degree in Environmental Management from Kyoto University. She has over a year of experience working as a consultant in the water sector with Ernst & Young LLP, India. She has also worked with various other organizations, including the United Nations Framework Convention on Climate Change (UNFCCC) and the Institute for Global Environmental Strategies (IGES), Japan.

Izuru Saizen is Professor at the Graduate School of Global Environmental Studies (GSGES), Kyoto University, with a concurrent appointment in the Graduate School of Agriculture. He holds a master's degree and PhD from Kyoto University. His research spans sustainable rural development, spatial planning, geographic information science (GIS), and regional environmental systems. He has conducted extensive fieldwork across Asia (e.g. -29- India, Vietnam, Laos, Cambodia) and integrates GIS and spatial analytics in his studies. He also engages in problem-solving research that addresses real-world issues in rural and urban environmental planning.